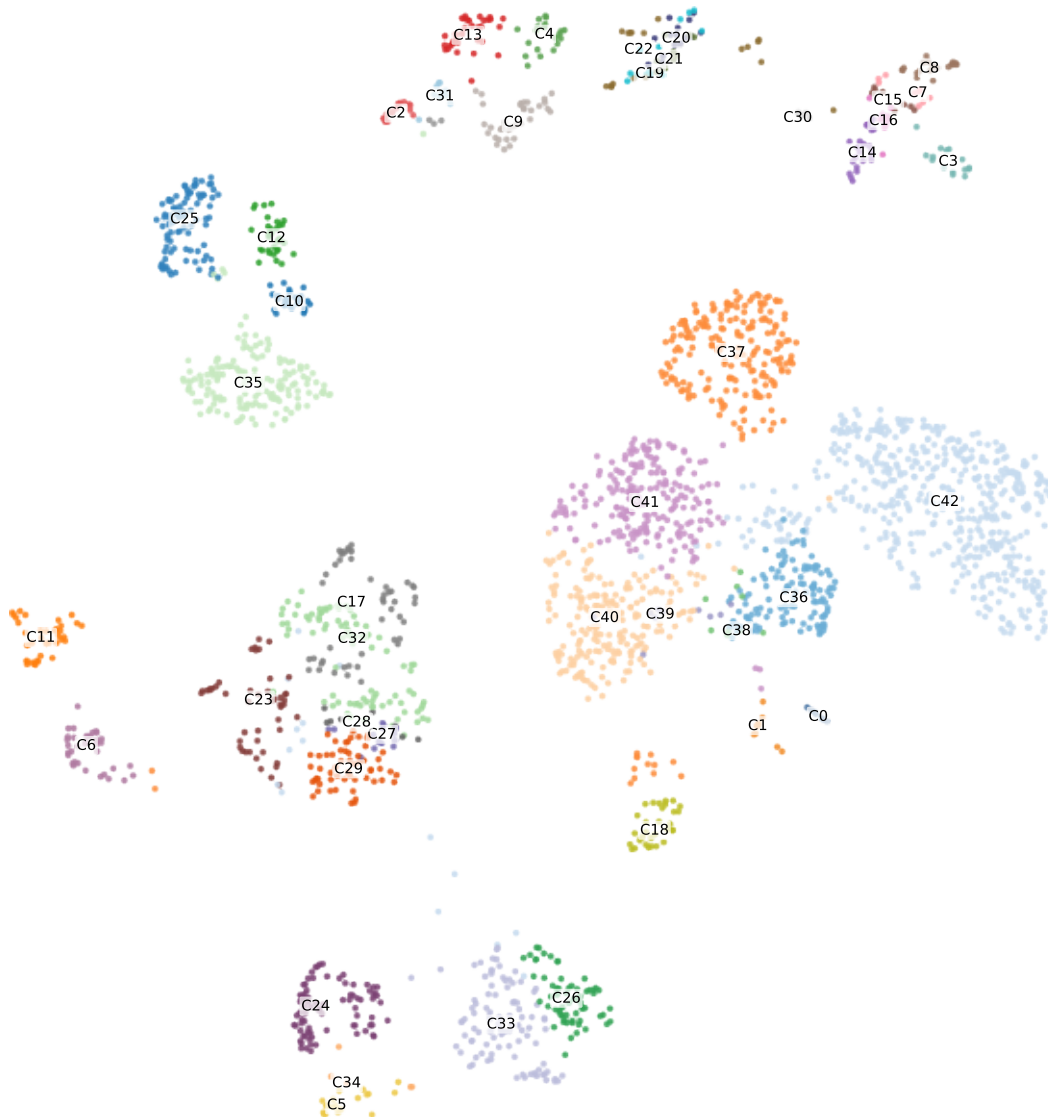
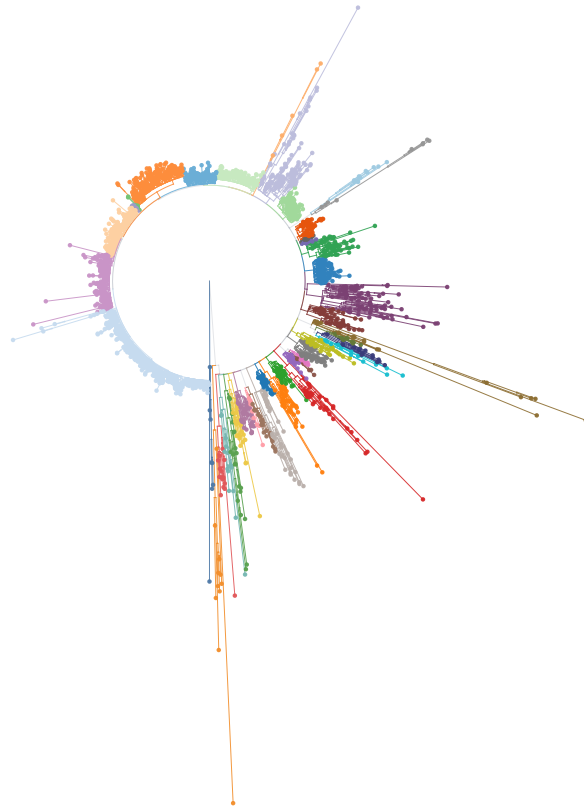


Adult pancreas scRNA benchmark selected NNLS-TBS clusters



Adult pancreas scRNA benchmark real full radial tree



All leaves are plotted; leaf colors match the selected-cluster UMAP.

adult: branch-length fit (50,000 sampled pairs)

cells / clusters: 2,500 / 43
sampled pairs: 50,000
R2: 0.974
correlation: 0.987
RMSE: 0.307; MAE: 0.201

NNLS tree path distance

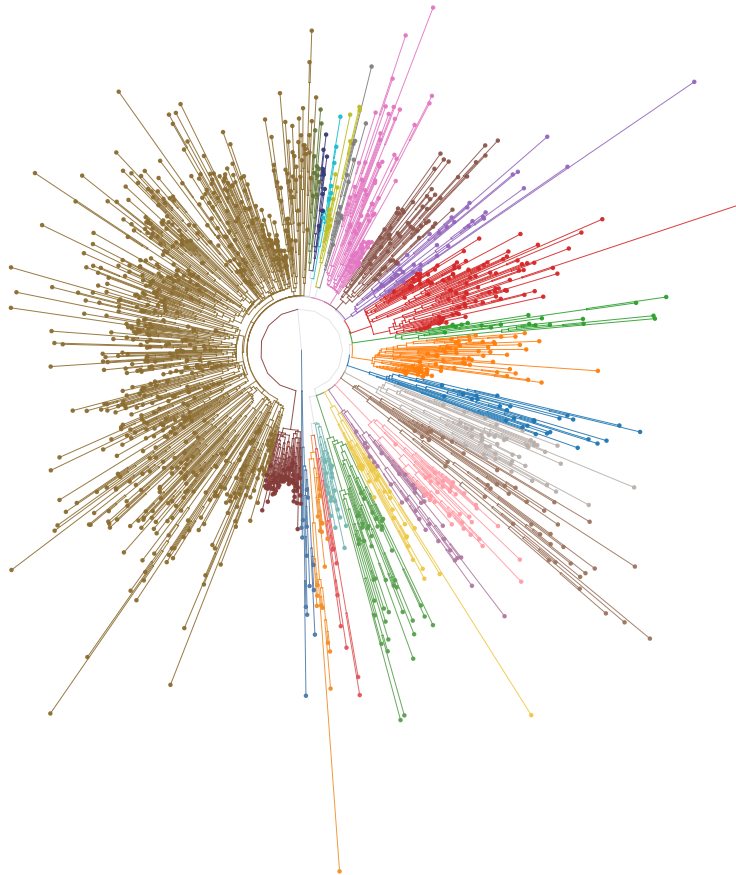
target pair distance

pair count, log scale

**Goncalves fetal pancreas progenitor benchmark
selected NNLS-TBS clusters**



Goncalves fetal pancreas progenitor benchmark real full radial tree



All leaves are plotted; leaf colors match the selected-cluster UMAP.

goncalves: branch-length fit (50,000 sampled pairs)

cells / clusters: 1,465 / 24
sampled pairs: 50,000
R2: 0.807
correlation: 0.899
RMSE: 0.335; MAE: 0.254

NNLS tree path distance

target pair distance

pair count, log scale

